

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I Shaik Mahaboob Basha (192365045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik

Mahaboob Basha

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \sim \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\sim$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk
- 4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\sim$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\sim$ Wumpus [x,y] $\wedge$ $\sim$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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1Q)

Given Statements

1. If a patient has Fever AND Cough, then the patient has Possible Flu.
2. If a patient has Fever AND Rash, then the patient has Possible Measles.
3. If a patient has Cough AND Breathlessness, then the patient has Possible Pneumonia.
4. Patient A has Fever.
5. Patient A has Cough.
6. Patient A has Breathlessness.

To Prove: Pneumonia(PatientA)

Step 1: Convert Facts into First-Order Logic (FOL)

1. $\forall x ((\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x))$
2. $\forall x ((\text{Fever}(x) \wedge \text{Rash}(x)) \rightarrow \text{Measles}(x))$
3. $\forall x ((\text{Cough}(x) \wedge \text{Breathlessness}(x)) \rightarrow \text{Pneumonia}(x))$
4. $\text{Fever}(\text{PatientA})$
5. $\text{Cough}(\text{PatientA})$
6. $\text{Breathlessness}(\text{PatientA})$

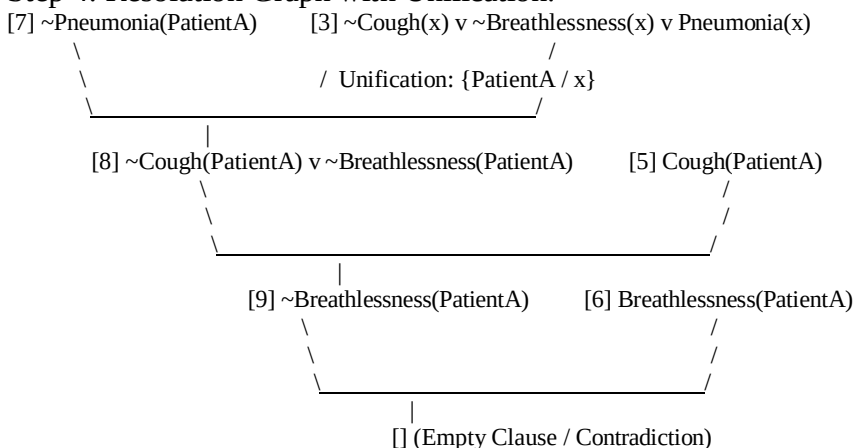
Step 2: Convert FOL into Conjunctive Normal Form (CNF)

1. $\sim \text{Fever}(x) \vee \sim \text{Cough}(x) \vee \text{Flu}(x)$
2. $\sim \text{Fever}(x) \vee \sim \text{Rash}(x) \vee \text{Measles}(x)$
3. $\sim \text{Cough}(x) \vee \sim \text{Breathlessness}(x) \vee \text{Pneumonia}(x)$
4. $\text{Fever}(\text{PatientA})$
5. $\text{Cough}(\text{PatientA})$
6. $\text{Breathlessness}(\text{PatientA})$

Step 3: Negate the Statement to be Proved

- Statement to prove: $\text{Pneumonia}(\text{PatientA})$
- Negated goal: $\sim \text{Pneumonia}(\text{PatientA})$

Step 4: Resolution Graph with Unification:



2Q)

Given Statements

1. If x is a vehicle and has an emergency type, then x has a clear path.
2. If x has a clear path, then x has a green signal and does not have a red signal.
3. If x is a vehicle, x is behind y , and y has a green signal, then x can proceed.
4. Ambulance is a vehicle.
5. Ambulance has an emergency type.
6. CarA is a vehicle.
7. CarA is behind Ambulance.

To Prove: Proceed(CarA)

Step 1: Convert Facts into First-Order Logic (FOL)

1. $\forall x ((\text{Vehicle}(x) \wedge \text{EmergencyType}(x)) \rightarrow \text{ClearPath}(x))$
2. $\forall x (\text{ClearPath}(x) \rightarrow (\text{GreenSignal}(x) \wedge \sim \text{RedSignal}(x)))$
3. $\forall x \forall y ((\text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y)) \rightarrow \text{Proceed}(x))$
4. $\text{Vehicle}(\text{Ambulance})$
5. $\text{EmergencyType}(\text{Ambulance})$
6. $\text{Vehicle}(\text{CarA})$
7. $\text{Behind}(\text{CarA}, \text{Ambulance})$

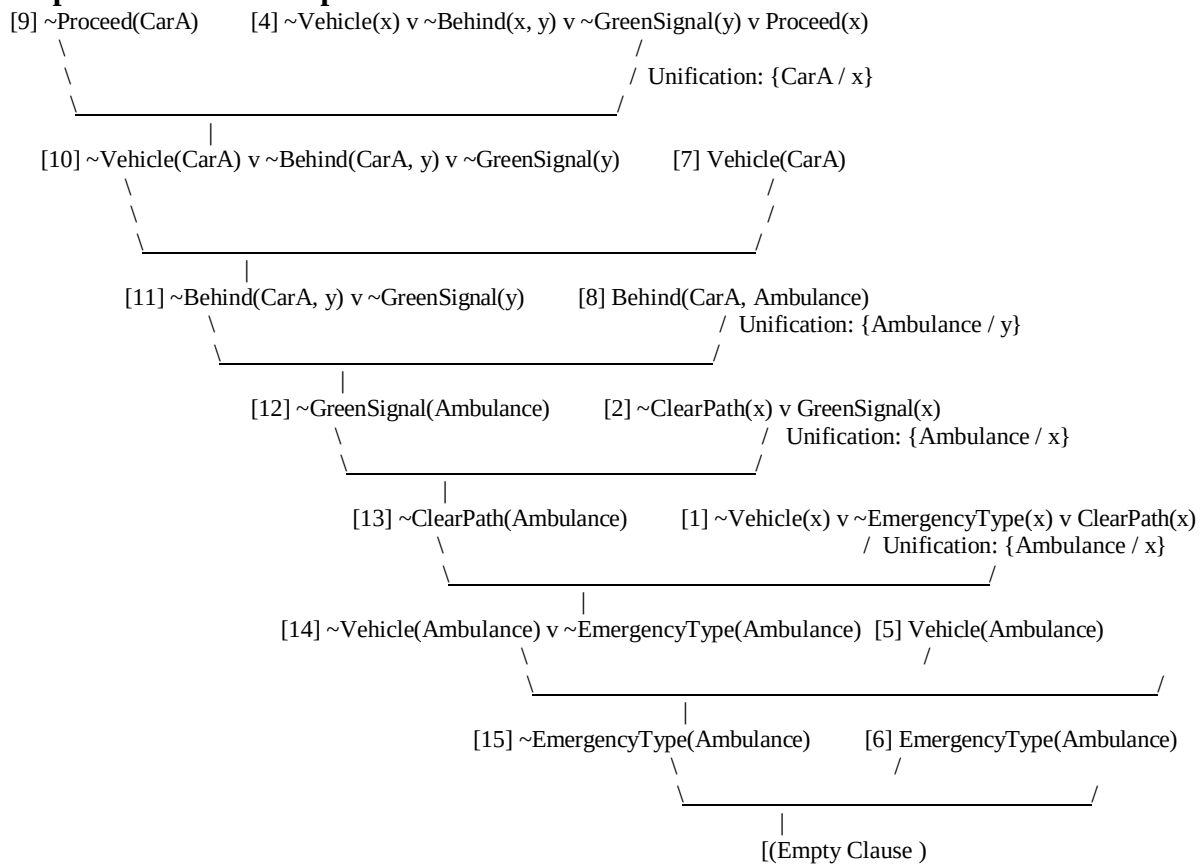
Step 2: Convert FOL into Conjunctive Normal Form (CNF)

1. $\sim \text{Vehicle}(x) \vee \sim \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
2. $\sim \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
3. $\sim \text{ClearPath}(x) \vee \sim \text{RedSignal}(x)$
4. $\sim \text{Vehicle}(x) \vee \sim \text{Behind}(x, y) \vee \sim \text{GreenSignal}(y) \vee \text{Proceed}(x)$
5. $\text{Vehicle}(\text{Ambulance})$
6. $\text{EmergencyType}(\text{Ambulance})$
7. $\text{Vehicle}(\text{CarA})$
8. $\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 3: Negate the Statement to be Proved

- Statement to prove: Proceed(CarA)
- Negated goal: $\sim \text{Proceed}(\text{CarA})$

Step 4: Resolution Graph with Unification



3Q)

Given Statements

1. If the soil is dry, then irrigation is needed.
2. If irrigation is needed and the crop is wheat, then apply the drip method.
3. If the drip method is not applied and the crop is wheat, then the crop is at risk.
4. The soil is dry.
5. The crop is wheat.

To Prove: Apply Drip Method

Step 1: Convert Facts into Propositional Logic

1. $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
2. $(\text{IrrigationNeeded} \wedge \text{CropWheat}) \rightarrow \text{ApplyDripMethod}$
3. $(\sim \text{ApplyDripMethod} \wedge \text{CropWheat}) \rightarrow \text{CropAtRisk}$
4. SoilDry
5. CropWheat

Step 2: Convert into Conjunctive Normal Form (CNF)

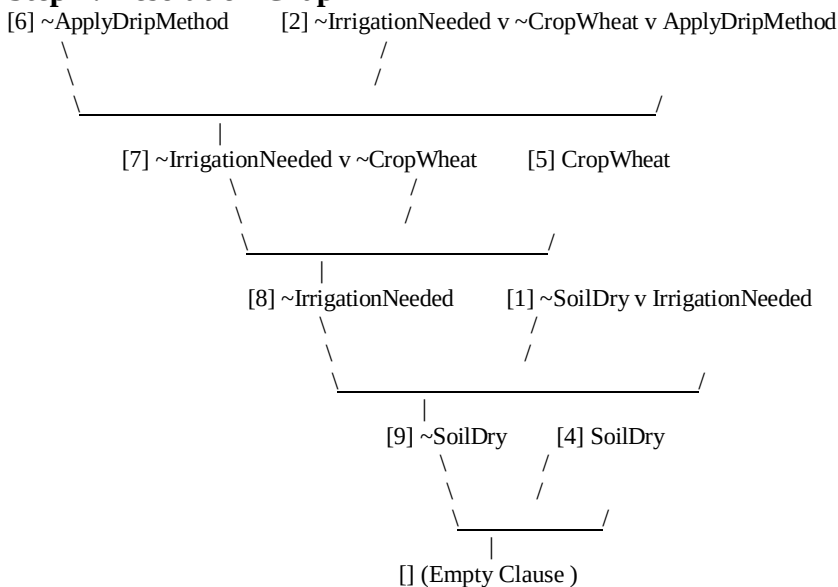
Eliminate implications ($\rightarrow$) by converting $A \rightarrow B$ to $\sim A \vee B$, and apply De Morgan's laws where necessary:

1. $\sim \text{SoilDry} \vee \text{IrrigationNeeded}$
2. $\sim \text{IrrigationNeeded} \vee \sim \text{CropWheat} \vee \text{ApplyDripMethod}$
3. $\text{ApplyDripMethod} \vee \sim \text{CropWheat} \vee \text{CropAtRisk}$ (Note: $\sim(\sim A \wedge C) \equiv A \vee \sim C$)
4. SoilDry
5. CropWheat

Step 3: Negate the Statement to be Proved

- Statement to prove: ApplyDripMethod
- Negated goal: $\sim \text{ApplyDripMethod}$

Step 4: Resolution Graph



(a) Belief State & Modus Ponens Derivations

1. Initial Belief State (Known Percepts & Base Facts)

- F_1 :Stench(1,2)
- F_2 :Breeze(1,1)
- F_3 :Glitter(2,2)
- F_4 : $\sim$ Wumpus(1,1) $\wedge$ $\sim$ Pit(1,1)(Standard start location assumption; [1,1] is known safe)

2. Systematic Application of Modus Ponens

$$\text{Modus Ponens Rule: } \frac{\alpha, \quad \alpha \rightarrow \beta}{\beta}$$

- **Inference 1:**
 - Premise: F_1 :Stench(1,2)
 - Rule 1: $\text{Stench}(1,2) \rightarrow \text{Wumpus}(1,1) \vee \text{Wumpus}(1,3) \vee \text{Wumpus}(2,2)$
 - Derivation: F_5 : $\text{Wumpus}(1,1) \vee \text{Wumpus}(1,3) \vee \text{Wumpus}(2,2)$
- **Inference 2:**
 - Premise: F_2 :Breeze(1,1)
 - Rule 2: $\text{Breeze}(1,1) \rightarrow \text{Pit}(1,2) \vee \text{Pit}(2,1)$
 - Derivation: F_6 : $\text{Pit}(1,2) \vee \text{Pit}(2,1)$
- **Inference 3:**
 - Premise: F_3 :Glitter(2,2)
 - Rule 3: $\text{Glitter}(2,2) \rightarrow \text{Gold}(2,2)$
 - Derivation: F_7 :Gold(2,2)

3. Derived Conclusions List

1. $\text{Wumpus}(1,1) \vee \text{Wumpus}(1,3) \vee \text{Wumpus}(2,2)$
2. $\text{Pit}(1,2) \vee \text{Pit}(2,1)$
3. Gold(2,2)

(b) Forward Chaining: Identifying Possible Locations

Using forward chaining and unit resolution to narrow down the hazard locations:

1. Wumpus Location Inference:

1. From F_5 , we have: $\text{Wumpus}(1,1) \vee \text{Wumpus}(1,3) \vee \text{Wumpus}(2,2)$.
2. From start cell knowledge (F_4), we know: $\sim \text{Wumpus}(1,1)$.
3. Applying **Unit Resolution** between $(\text{Wumpus}(1,1) \vee \text{Wumpus}(1,3) \vee \text{Wumpus}(2,2))$ and $\sim \text{Wumpus}(1,1)$:

$$\text{Wumpus}(1,3) \vee \text{Wumpus}(2,2)$$

4. **Conclusion:** The Wumpus is located at [1,3] or [2,2].

2. Pit Location Inference:

1. From F_6 , we have: $\text{Pit}(1,2) \vee \text{Pit}(2,1)$.
2. Since no further percepts (such as $\sim$ Breeze at other surrounding cells) are provided, neither $\text{Pit}(1,2)$ nor $\text{Pit}(2,1)$ can be disproved.
3. **Conclusion:** A Pit exists at [1,2], [2,1], or **both**.

(c) Path Safety Evaluation

Target Path: $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

To prove a cell $[x,y]$ is safe using Rule 4 ($\sim Wumpus(x,y) \wedge \sim Pit(x,y) \rightarrow Safe(x,y)$), both $\sim Wumpus(x,y)$ and $\sim Pit(x,y)$ must be logically derived.

Step 1: Evaluation of $[1,1]$

- **Status:** SAFE
- **Inference:** True by initial condition F_4 .

Step 2: Evaluation of $[1,2]$

- Wumpus Check: No stench at $[1,1]$ implies $\sim Wumpus(1,2)$.
- Pit Check: From F_6 , $Pit(1,2) \vee Pit(2,1)$ is true. Thus, $\sim Pit(1,2)$ cannot be proved.
- Safety Evaluation: Since $\sim Pit(1,2)$ is unproven, Rule 4 cannot fire for cell $[1,2]$.
- Status: UNSAFE / HIGH RISK (Cell $[1,2]$ may contain a Pit).

Step 3: Evaluation of $[2,2]$

- Gold Check: Gold(2,2) is confirmed from F_7 .
- Wumpus Check: From section (b), $Wumpus(1,3) \vee Wumpus(2,2)$ is true. Thus, $\sim Wumpus(2,2)$ cannot be proved.
- Safety Evaluation: Since $\sim Wumpus(2,2)$ is unproven, Rule 4 cannot fire for cell $[2,2]$.
- Status: UNSAFE / HIGH RISK (Cell $[2,2]$ may contain the Wumpus).

Conclusion

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is NOT SAFE.

- Traversing to $[1,2]$ risks falling into a Pit.
- Traversing to $[2,2]$ risks encountering the Wumpus.